

My Notebook

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Ariel Smyrichinsky (arismyri@iu.edu)

on

Sep 15, 2026 @12:12 PM EDT

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Ariel Smyrichinsky (arismyri@iu.edu) - Sep 08, 2026, 1:10 PM EDT

12:58 PM **Ariel Smyrichinsky**, Camden Bush, Garrett Willis

Ariel Smyrichinsky - Sep 08, 2026, 12:48 PM EDT

I have downloaded the Jupyter notebook for this lab. The file name is P451_A_EM_Pulses_2026_Jun_10.ipynb, but I plan on making my own copy and renaming it later if possible.

The goal of this lab is to investigate the properties of coaxial cables. Coaxial cables are prevalent in many lab settings because of their efficiency in transmitting signals without information loss and lesser chance of interference. Coaxial cables used in a lab setting tend to have a BNC (Berkeley Nucleonics Corporation) connector at the end of it, which makes for added security due to its twisting and locking fashion.

The procedure of this lab is as follows:

1. Measure the signal propagation velocity for one of the cables, and note the time delay and physical length of the cable. Also consider if the methodology for how the time is determined matters. What do the differences between the incident and reflected pulses mean? Do the same for both cables.
2. Determine what the probability is that the two cables will have a velocity difference at least as great as the one that we measured?
3. Then, we will terminate the coaxial cable with a variety of different impedances, and study the ratio of reflected to incident pulses as a function of load impedance.
4. Make a plot of reflection coefficient as a function of the load impedance, then fit the data to the expected theoretical form. Does the fit match expectations? How is the quality of the fit?
5. Modify the fit to include the effects of attenuation in the cable. What changes? What improves?

Ariel Smyrichinsky (arismyri@iu.edu) - Sep 08, 2026, 3:00 PM EDT

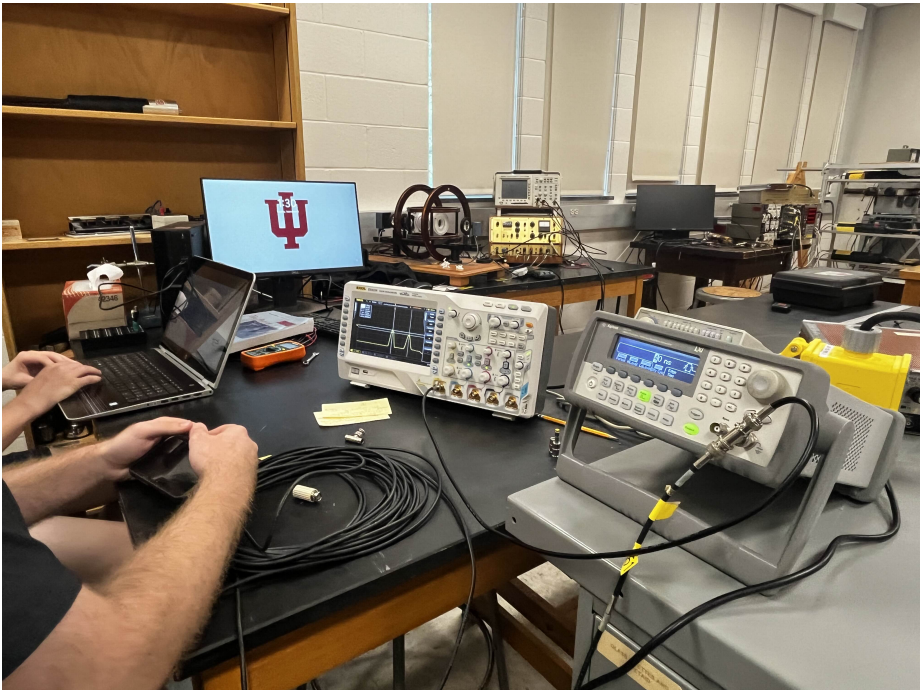
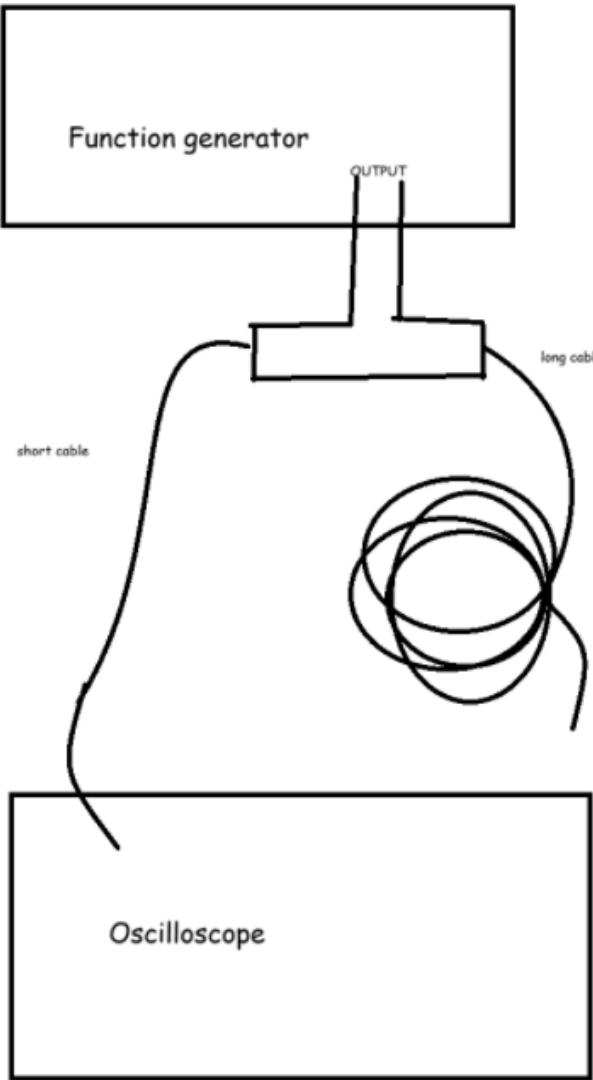
1:10 PM **Ariel Smyrichinsky**, Camden Bush, Garrett Willis

Willis has attached the oscilloscope (RIGOL DS4034 digital oscilloscope, ultravision, 4 channel 350 MHz, 4 GS/s) to the function generator (Agilent 33210A 10 MHz) with a coaxial cable and is examining the system.

1:19 PM

We have a T connector attached to the function generator. Attached to the T connector is a shorter cable and a longer cable. The shorter cable is connected to the oscilloscope, and the longer cable is not connected to anything at the other end.

When a pulse is sent from the function generator, it will propagate through both cables. Because there is nothing connected at the end of the longer cable, the pulse that goes down that way will reflect, and will show up as a second peak that follows the peak that goes through the cable connected directly to the oscilloscope. This is due to the time delay from the signal propagating.



1:40 PM

Notably, the BNC connector affixed to the oscilloscope got somewhat warm.

1:51 PM

$$(\text{total length of wires}) = (\text{signal propagation})(\text{time delay between pulses})$$

We are connecting cable 1 and cable 3 to the function generator; cable 1 is not connected at the other end, and cable 3 is connected to the oscilloscope.

The function generator is generating a pulse of 40 ns width, 20 ns edge time (rise and fall), 100 mv peak to peak amplitude. And there you go.

(total length of wires) = (signal propagation velocity)(time delay between pulses)

$2(15.1765 \pm 0.004 \text{ meters}) + (0.97155 \pm 0.004 \text{ meters}) = (\text{signal propagation velocity})(153 \text{ nanoseconds} \pm 5 \text{ nanosecond})$

$(\text{signal propagation velocity}) = (31.325 \pm 0.009 \text{ meters}) / (0.000000153 \text{ nanoseconds} \pm 0.000000005 \text{ nanosecond})$

(signal propagation velocity) = $205000000 \pm 7000000 \text{ meters/second} = 0.683c \pm 0.02c$ where $c = 3e8 \text{ meters/second}$.

When compared to the datasheets in the lab writeup, this is consistent with the specified 66c.

8259 Coax - RG-58A

RG Type:	58A/U
Suitability	
Suitability - Indoor:	Yes
Suitability - Outdoor:	Yes
Suitability - Aerial:	Yes - when supported by a messenger wire
Plenum/Non-Plenum	
Plenum (Y/N):	No

Electrical Characteristics (Overall)

Nom. Characteristic Impedance:

Impedance (Ohm)
50

Nom. Inductance:

Inductance (pH/m)
0.301852

Nom. Capacitance Conductor to Shield:

Capacitance (pF/m)
101.055

Nominal Velocity of Propagation:

VP (%)
66

Nominal Delay:

Delay (ns/m)
5.05274

Nom. Conductor DC Resistance:

DCR @ 20°C (Ohm/km)
35.4348

Nominal Outer Shield DC Resistance:

DCR @ 20°C (Ohm/km)
13.4521

Nom. Attenuation:

Freq. (MHz)	Attenuation (dB/100m)
1	1.3124
10	4.9215
50	12.1397
100	17.7174
200	26.5761
400	40.6844
700	58.0737
900	69.2291
1000	74.8068

Max. Operating Voltage - Non-UL:

2:50 PM

I have removed cable 1. Bush has introduced cable 2 to the system.

(total length of wires) = (signal propagation velocity)(time delay between pulses)

$2(12.1666 \pm 0.004 \text{ meters}) + (0.97155 \pm 0.004 \text{ meters}) = (\text{signal propagation velocity})(113 \text{ nanoseconds} \pm 5 \text{ nanosecond})$

$(\text{signal propagation velocity}) = (25.305 \pm 0.009 \text{ meters}) / (0.000000113 \text{ nanoseconds} \pm 0.000000005 \text{ nanosecond})$

$(\text{signal propagation velocity}) = 220000000 \pm 10000000 \text{ meters/second} = 0.733c \pm 0.03c$ where $c = 3e8 \text{ meters/second}$.

When compared to the datasheets in the lab writeup, this is consistent with the specified 73c.

Detailed Specifications & Technical Data



METRIC MEASUREMENT VERSION

8219 Coax - RG-58A/U Type

EU Directive 2003/11/EC (BFR):	Yes
CA Prop 65 (CJ for Wire & Cable):	Yes
MIL Order #39 (China RoHS):	Yes
PMSHA Specification:	SC-182/5
RG Type:	58A/U
Flame Test	
UL Flame Test:	UL1685 UL Loading
Suitability	
Suitability - Indoor:	Yes
Suitability - Outdoor:	Yes - Black jacket only
Suitability - Aerial:	Yes - Black jacket only - when supported by a messenger wire
Plenum/Non-Plenum	
Plenum (Y/N):	No

Electrical Characteristics (Overall)

Nom. Characteristic Impedance:

Impedance (Ohm)

54

Nom. Inductance:

Inductance (µH/m)

0.213265

Nom. Capacitance Conductor to Shield:

Capacitance (pF/m)

86.9465

Nominal Velocity of Propagation:

VP (%)

73

Nominal Delay:

Delay (ns/m)

4.56059

Ariel Smyrichinsky (arismyri@iu.edu) - Sep 14, 2026, 8:49 PM EDT

3:01 PM

Next, we are going to reconnect cable 1 and terminate it with a variety of load impedances Z_L , with the goal of studying the ratio of reflected to incident pulses as a function of load impedance.

$$\Gamma = \frac{Z_L - Z_c}{Z_L + Z_c},$$

The coefficient of reflection \Gamma is given by

where Z_L is the load input impedance and Z_c is the cable impedance.

For changing the resistance, we are using an Elenco model RS-500 resistance substitution box.

	Table 1	
Resistance value [ohms]	Peak 1 height [mv]	Peak height 2 [mV]
0	102.6 +/- 2.0 mv	-93.24 +/- 2.0 mV
10	102.6 +/- 2.0 mv	-66.12 +/- 2.0 mV
20	102.6 +/- 2.0 mv	-46.0 +/- 2 mV
30	102.6 +/- 2.0 mv	-31.16 +/- 1.5 mV
40	102.6 +/- 2.0 mv	-19.76 +/- 1.5 mV

We have determined that resistance = 50 is not a good data point because that is close to the impedance matching point, and it is hard to discern where the peak is.

	Table 1 cont.	
100	101.8 +/- 2.0	25 +/- 1.5
200	101.8 +/- 2.0	50 +/- 1.0
500	101.8 +/- 2.0	69.16 +/- 3.0
1000	101.8 +/- 1.4	77.52 +/- 1.5
2000	101.8 +/- 2.5	81.32 +/- 1.5
3000	101.8 +/- 1.5	83.6 +/- 1.5
4000	103.0 +/- 2.0	83.6 +/- 1.3
5000	103.0 +/- 2.0	84.5 +/- 0.75
10000	103.0 +/- 2.0	84 +/- 1.5

3:54 PM

Switched out cable to cable 2.

	Table 2	
Resistance [ohms]	Incident wave [mV]	Reflected wave [mV]
0	102.0 +/- 2	-94 +/- 2
10	102.0 +/- 2	-66.5 +/- 2
20	102.0 +/- 2	-47.0 +/- 2
30	102.0 +/- 2	-32.0 +/- 2
40	102.0 +/- 2	-22.0 +/- 2
10k	102.0 +/- 2	87.0 +/- 2

	Table 2	
5k	102.0 +/- 2	86.0 +/- 2
4k	102.0 +/- 2	86.0 +/- 2
3k	102.0 +/- 2	85.0 +/-2
2k	102.0 +/- 2	83.0 +/- 2
1k	102.0 +/- 2	80.0 +/- 2
500	102.0 +/- 2	71.0 +/- 2
200	102.0 +/- 2	50.0 +/-2
100	102.0 +/- 2	26.5 +/- 1.5

Ariel Smyrichinsky (arismyri@iu.edu) - Sep 14, 2026, 8:27 PM EDT

4:20 PM

We have the resistance source hooked up to an Agilent 34401A 6 1/2 digit multimeter. The multiple wires are to account for the wire resistance.

Table 3	
Supposed resistance	True resistance
10	10.510+/- 0.005
20	20.380+/- 0.002
30	30.351+/- 0.001
40	40.476+/- 0.001
100	99.971+/- 0.003
200	199.60+/- 0.005
500	498.04+/- 0.005
1000	996.99+/- 0.005
2000	1999.9+/- 0.005
3000	2990.0+/- 0.005
4000	3995.8+/- 0.005
5000	4992.4+/- 0.005
10000	10982+/- 0.005

Ariel Smyrichinsky (arismyri@iu.edu) - Sep 14, 2026, 10:45 PM EDT

6:18 PM Ariel Smyrichinsky

I am going through the Jupyter notebook and altering the code to verify that I fully understand what it is doing.

8:04 PM

This is how the modified Jupyter notebook works:

The first code block imports packages that are necessary.

```
1 import numpy as np
2 import pandas as pd
3 import matplotlib.pyplot as plt
4 from scipy.optimize import curve_fit
```

numpy is a package that assists with array manipulation and mathematics.

pandas is a package that allows you to easily import datasets and analyze them.

matplotlib.pyplot is a package that allows you to intuitively make graphs of data.

The package scipy.optimize has the prewritten function curve_fit which allows you to perform fits.

The second code block defines the function that contains the equation we want to fit the data to, named "fit_function". The equation is the one mentioned above that describes the coefficient of reflection \Gamma.

$$\Gamma = \frac{Z_L - Z_c}{Z_L + Z_c},$$

where Z_L and Z_c are load input impedance and cable impedance, respectively.

```
1 # describes reflection from a cable of impedance z_c with load z_l
2
3 def fit_function(z_l, *popt): # the asterisk collects all the passed parameters into a tuple
4     z_c = popt[0]
5     return (z_l - z_c)/(z_l+z_c)
6
```

The third code block takes the data for cable 1 (which is stored in the text file "smyrichinsky_cable_1.txt" and contains the information in table 1 above) and imports it as a pandas dataframe named df_cable_1.

Then, the code block takes the columns in df_cable_1 and turns them to numpy arrays with the function .to_numpy().

The reflection coefficient and its error is then calculated for each coefficient and stored in gamma_1 and gamma_error_1 respectively.

```
1 df_cable_1 = pd.read_csv("smyrichinsky_cable_1.txt")
2
3 # columns: resistance, incident_wave_ampl, incident_wave_error, reflected_wave_ampl, reflected_wave_error
4
5 print(list(df_cable_1))
6
7 resistance_1 = df_cable_1["resistance"].to_numpy()
8 incident_ampl_1 = df_cable_1["incident_wave_ampl"].to_numpy()
9 incident_error_1 = df_cable_1["incident_wave_error"].to_numpy()
10 reflected_ampl_1 = df_cable_1["reflected_wave_ampl"].to_numpy()
11 reflected_error_1 = df_cable_1["reflected_wave_error"].to_numpy()
12
13 gamma_1 = reflected_ampl_1/incident_ampl_1
14 gamma_error_1 = np.abs(gamma_1)*np.sqrt( (reflected_error_1/reflected_ampl_1)**2 + (incident_error_1/incident_ampl_1)**2 )
15 print(gamma_1)
```

	Cable 1			
Resistance value [ohms]	Peak 1 height [mv]	Peak height 2 [mV]	Reflection coefficient	Reflection coefficient error
0	102.6 +/- 2.0 mv	-93.24 +/- 2.0 mV	-0.914	0.026
10	102.6 +/- 2.0 mv	-66.12 +/- 2.0 mV	-0.648	0.1023
20	102.6 +/- 2.0 mv	-46.0 +/- 2 mV	-0.460	0.021
30	102.6 +/- 2.0 mv	-31.16 +/- 1.5 mV	-0.305	0.020
40	102.6 +/- 2.0 mv	-19.76 +/- 1.5 mV	-0.193	0.019
100	101.8 +/- 2.0	25 +/- 1.5	0.245	0.015
200	101.8 +/- 2.0	50 +/- 1.0	0.490	0.013
500	101.8 +/- 2.0	69.16 +/- 3.0	0.678	0.032
1000	101.8 +/- 1.4	77.52 +/- 1.5	0.760	0.020
2000	101.8 +/- 2.5	81.32 +/- 1.5	0.797	0.021
3000	101.8 +/- 1.5	83.6 +/- 1.5	0.819	0.021
4000	103.0 +/- 2.0	83.6 +/- 1.3	0.819	0.020
5000	103.0 +/- 2.0	84.5 +/- 0.75	0.828	0.017
10000	103.0 +/- 2.0	84 +/- 1.5	0.823	0.021

The fourth code block applies the fit to the data to get a value for Z_c , the load input impedance.

The best fit it found was a load input impedance of 60.81 ± 1.16 ohms. This fit had a chi-squared of $186.02/13 = 14$.

```

1  guess = [50]
2
3  best_fit, covariance = curve_fit(fit_function, resistance_1, gamma_1, sigma = gamma_error_1,
4  | | | | | | | | | | p0 = guess, absolute_sigma = True)
5
6  perr = np.sqrt(np.diag(covariance))
7  fit = fit_function(resistance, *best_fit)
8  chi_squared = np.sum(((gamma_1 - fit)/gamma_error_1)**2)
9  ndof = len(gamma_1) - len(guess)
10
11 print("best fit = %f +- %f ohms"%(best_fit[0],perr[0]))
12 print('chisq = %f / %d'%(chi_squared,ndof))
✓ 0.0s

```

best fit = 60.814857 +- 1.167440 ohms

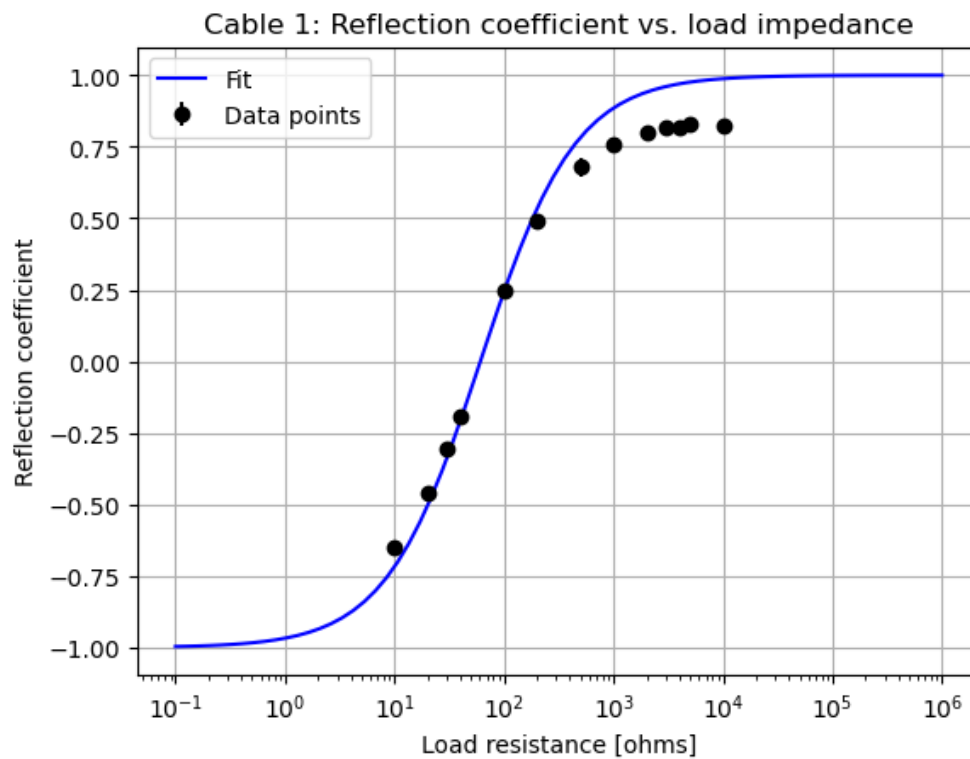
chisq = 186.028494 / 13

The fifth code block plots the fit against the data.

```

1  fig, ax = plt.subplots()
2  ax.errorbar(resistance_2, gamma_2, yerr = gamma_error_2, fmt = 'o', color = "black", label = "Data points")
3  rl_smooth = 10**(np.linspace(-1,6,70))
4  best_fit_smooth = fit_function(rl_smooth,*best_fit)
5  ax.plot(rl_smooth, best_fit_smooth, color = "blue", label = "Fit")
6  ax.set_xscale("log")
7  ax.grid()
8  ax.set_xlabel("Load resistance [ohms]")
9  ax.set_ylabel("Reflection coefficient")
10 ax.set_title("Cable 1: Reflection coefficient vs. load impedance")
11 ax.legend()

```



The fit is not ideal.

Ariel Smyrichinsky (arismyri@iu.edu) - Sep 14, 2026, 10:54 PM EDT

8:46 PM

The following code blocks perform a similar process for the second cable, using the data from table 2, stored in the text file "smyrichinsky_cable_2.txt." Variables have been denoted with a 2 instead of a 1 to show that they are values for cable 2.

```

1 df_cable_2 = pd.read_csv("smyrichinsky_cable_2.txt")
2
3 # columns: resistance, incident_wave_ampl, incident_wave_error, reflected_wave_ampl, reflected_wave_error
4
5 print(list(df_cable_2))
6
7 resistance_2 = df_cable_2["resistance"].to_numpy()
8 incident_ampl_2 = df_cable_2["incident_wave_ampl"].to_numpy()
9 incident_error_2 = df_cable_2["incident_wave_error"].to_numpy()
10 reflected_ampl_2 = df_cable_2["reflected_wave_ampl"].to_numpy()
11 reflected_error_2 = df_cable_2["reflected_wave_error"].to_numpy()
12
13 gamma_2 = reflected_ampl_2/incident_ampl_2
14 gamma_error_2 = np.abs(gamma_2)*np.sqrt( (reflected_error_2/reflected_ampl_2)**2 + (incident_error_2/incident_ampl_2)**2 )
15 print(gamma_2)

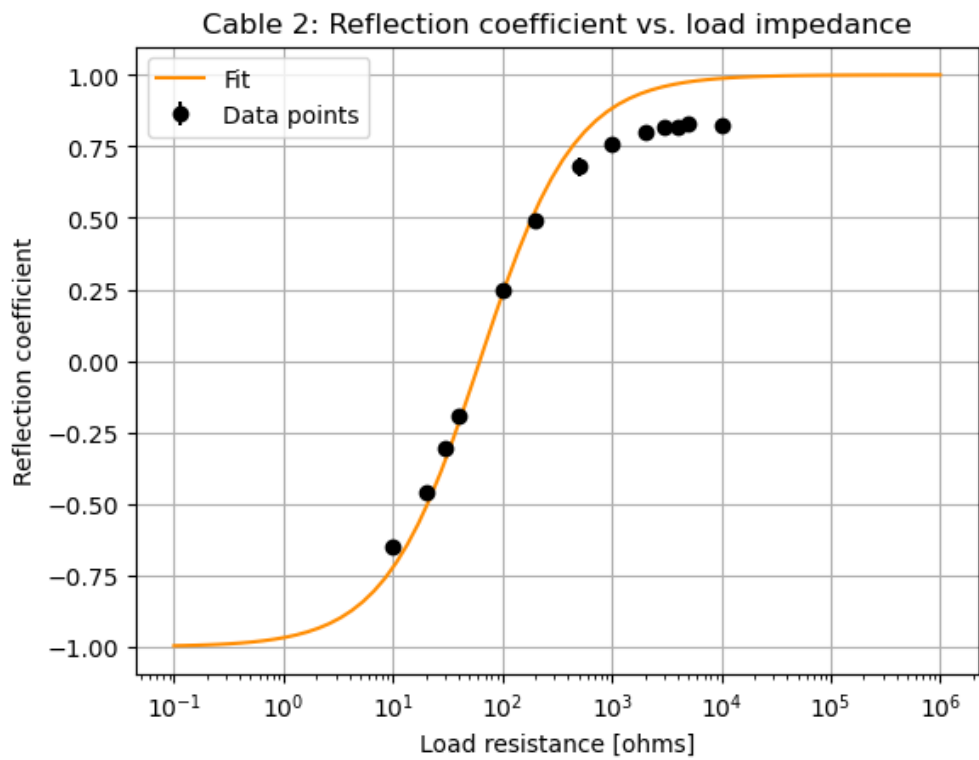
```

```
1 guess = [50]
2
3 best_fit, covariance = curve_fit(fit_function, resistance_2, gamma_2, sigma = gamma_error_2,
4 | | | | | | | | | | p0 = guess, absolute_sigma = True)
5
6 perr = np.sqrt(np.diag(covariance))
7 fit = fit_function(resistance_2, *best_fit)
8 chi_squared = np.sum(((gamma_2 - fit)/gamma_error_2)**2)
9 ndof = len(gamma_2) - len(guess)
10
11 print("best fit = %f +- %f ohms"%(best_fit[0],perr[0]))
12 print('chisq = %f / %d'%(chi_squared,ndof))
```

```
1 fig, ax = plt.subplots()
2 ax.errorbar(resistance, gamma, yerr = gamma_error, fmt = 'o', color = "black", label = "Data points")
3 rl_smooth = 10**(np.linspace(-1,6,70))
4 best_fit_smooth = fit_function(rl_smooth,*best_fit)
5 ax.plot(rl_smooth, best_fit_smooth, color = "darkorange", label = "Fit")
6 ax.set_xscale("log")
7 ax.grid()
8 ax.set_xlabel("Load resistance [ohms]")
9 ax.set_ylabel("Reflection coefficient")
10 ax.set_title("Cable 2: Reflection coefficient vs. load impedance")
11 ax.legend()
```

The best fit it found was a load input impedance of 62.24 +- 1.11 ohms.

This fit had a chi-squared of 346.917726 / 13 = 26.68.



	Cable 2			
Resistance [ohms]	Incident wave [mV]	Reflected wave [mV]		
0	102.0 +/- 2	-94 +/- 2	-0.921	0.026

	Cable 2			
10	102.0 +/- 2	-66.5 +/- 2	-0.651	0.023
20	102.0 +/- 2	-47.0 +/- 2	-0.461	0.021
30	102.0 +/- 2	-32.0 +/- 2	-0.313	0.020
40	102.0 +/- 2	-22.0 +/- 2	-0.215	0.020
10k	102.0 +/- 2	87.0 +/- 2	0.853	0.025
5k	102.0 +/- 2	86.0 +/- 2	0.843	0.025
4k	102.0 +/- 2	86.0 +/- 2	0.843	0.025
3k	102.0 +/- 2	85.0 +/- 2	0.833	0.025
2k	102.0 +/- 2	83.0 +/- 2	0.813	0.025
1k	102.0 +/- 2	80.0 +/- 2	0.784	0.024
500	102.0 +/- 2	71.0 +/- 2	0.696	0.023
200	102.0 +/- 2	50.0 +/- 2	0.490	0.218
100	102.0 +/- 2	26.5 +/- 1.5	0.259	0.015

Ariel Smyrichinsky (arismyri@iu.edu) - Sep 14, 2026, 10:54 PM EDT

10:15 PM

1. The signal propagation velocity for cable A (model ?) was $0.68c \pm 0.02c$, where c is $3e8$, the speed of light. The signal propagation velocity for cable B was $0.73c \pm 0.03c$. In both cases, there was a difference between the incident and reflected pulses: the reflected pulse had a smaller amplitude when compared to the incident pulse. This is likely due to loss of energy as the pulse propagates down the wire.

2. $V_B - V_A = 0.05c \pm 0.04c$. I believe the probability of two same cables to have a velocity difference deviation as large as the error on $V_B - V_A$ is quite low.

3. (Peak and pulse are interchangeable here.) The load impedance at zero gives the largest absolute value ratio between the pulse heights. For cable one, the ratio of the reflected to incident pulse is $(-93.24 \pm 2.0) / (102.6 \pm 2.0) = -0.91 \pm 0.03$. For cable two, the ratio of the reflected to incident pulse is $(-94 \pm 2) / (102.0 \pm 2) = -0.92 \pm 0.03$.

As the load impedance increases to 50, the ratio between the pulse heights decreases, until it reaches zero (the impedance matching point.) Once the load impedance increases past 50, the ratio begins to increase again.

The highest load impedance we terminated the cables with was 10000 ohms. At this resistance, the ratio of the reflected to incident pulse for cable one is $(84 \pm 1.5) / (103.0 \pm 2.0) = 0.82 \pm 0.02$. The ratio of reflected to incident pulse for cable two is $(87.0 \pm 2.0) / (102.0 \pm 2.0) = 0.85 \pm 0.03$

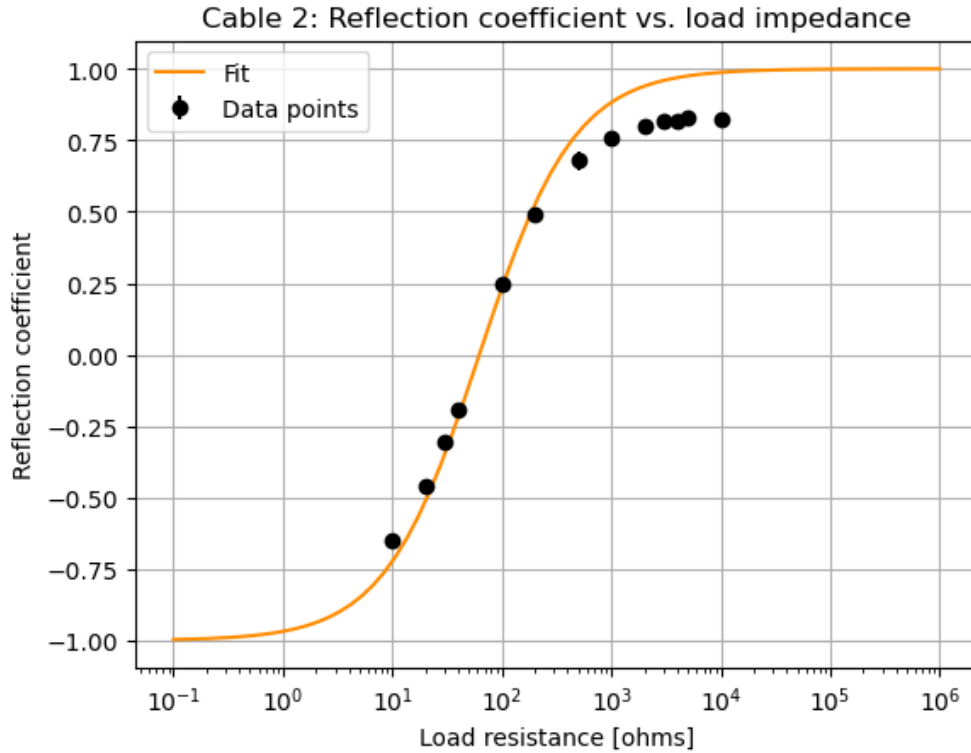
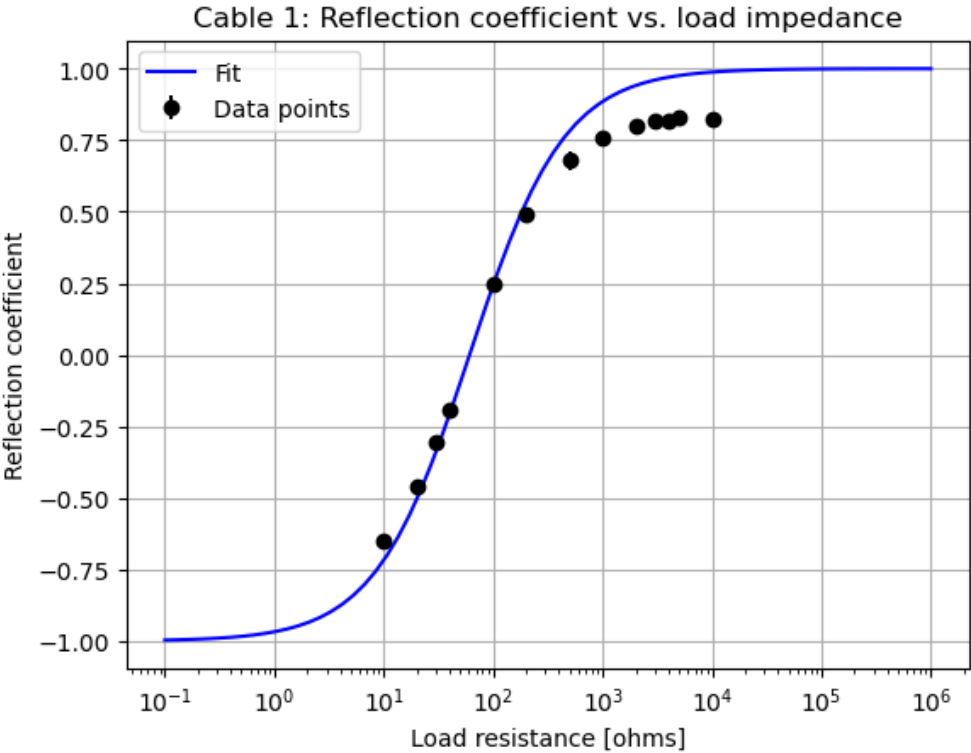
	Cable 1	
Resistance value [ohms]	Peak 1 height [mv]	Peak height 2 [mV]
0	102.6 +/- 2.0 mv	-93.24 +/- 2.0 mV
10	102.6 +/- 2.0 mv	-66.12 +/- 2.0 mV
20	102.6 +/- 2.0 mv	-46.0 +/- 2 mV
30	102.6 +/- 2.0 mv	-31.16 +/- 1.5 mV
40	102.6 +/- 2.0 mv	-19.76 +/- 1.5 mV
100	101.8 +/- 2.0	25 +/- 1.5
200	101.8 +/- 2.0	50 +/- 1.0
500	101.8 +/- 2.0	69.16 +/- 3.0
1000	101.8 +/- 1.4	77.52 +/- 1.5
2000	101.8 +/- 2.5	81.32 +/- 1.5
3000	101.8 +/- 1.5	83.6 +/- 1.5
4000	103.0 +/- 2.0	83.6 +/- 1.3
5000	103.0 +/- 2.0	84.5 +/- 0.75
10000	103.0 +/- 2.0	84 +/- 1.5

	Cable 2	
Resistance [ohms]	Incident wave [mV]	Reflected wave [mV]
0	102.0 +/- 2	-94 +/- 2
10	102.0 +/- 2	-66.5 +/- 2
20	102.0 +/- 2	-47.0 +/- 2
30	102.0 +/- 2	-32.0 +/- 2
40	102.0 +/- 2	-22.0 +/- 2
10k	102.0 +/- 2	87.0 +/- 2
5k	102.0 +/- 2	86.0 +/- 2
4k	102.0 +/- 2	86.0 +/- 2
3k	102.0 +/- 2	85.0 +/- 2
2k	102.0 +/- 2	83.0 +/- 2
1k	102.0 +/- 2	80.0 +/- 2
500	102.0 +/- 2	71.0 +/- 2
200	102.0 +/- 2	50.0 +/- 2
100	102.0 +/- 2	26.5 +/- 1.5

4. The fit values don't quite match my expectation. I expected the impedance matching point to be 50 for both.

Cable 1: the Z_c fit value was 60.81 +/- 1.16 ohms.

Cable 2: the Z_c fit value was 62.24 +/- 1.11 ohms.



5. Via visual inspection, the fit quality is not the best. For cable 1, the chi-squared/dof is $186.02/13 = 14$. For cable 2, the chi-squared/dof is $346.917726 / 13 = 26.68$.